MALLA REDDY UNIVERSITY

IV Year B Tech I Semester

L/T/P/C
3/0/0/3

(MR20-1CS0307) BIG DATA ANALYTICS AND APPLICATIONS

COURSE OBJECTIVES

1. To know the fundamental concepts of big data.
2. Realize the Hadoop architecture and implementation of MapReduce Application.
3. Understand Big Data technologies and NoSQL.
4. To explore tools and practices for working with big data technologies.
5. To know the statistical programing patterns in scala.

UNIT I:

Introduction to Big Data: Defining Big Data, Big Data Types, Analytics, examples, Technologies, The evolution of Big Data Architecture.

Basics of Hadoop: Hadoop Architecture, Main Components of Hadoop Framework, Analysis Big data using Hadoop, Hadoop clustering.

UNIT-II:

MapReduce: A Weather DataSet, Analyzing the data with Unix Tool, Analyzing the data using Hadoop, Hadoop streaming, Hadoop Pipes.

The Hadoop Distributed File System: Design of HDFS, Concepts, Basic File system Operations, Interfaces, Data Flow.

Hadoop I/O: Data Integrity, Compression, Serialization, File-Based Data Structures.

UNIT-III:

Developing A MapReduce Application: UNIT Tests with MRUNIT, Running Locally on Test Data.

How MapReduce Works: Anatomy of MapReduce Job Run, Classic MapReduce, Yarn, Failures in Classic MapReduce and Yarn, Job Scheduling, Shuffle and Sort, Task Execution.

MapReduce Types and Formats: MapReduce types, Input Formats, Output Formats.

UNIT –IV:

NoSQL Data Management: Introduction, Types of NoSQL, Query Model for Big Data, Benefits of NoSQL, MongoDB.

Hbase: Data Model and Implementations, Hbase Clients, Hbase Examples, Praxis.

Hive: Comparison with Traditional Databases, HiveQL, Tables, Querying Data, User Defined Functions.

Sqoop: Sqoop Connectors, Text and Binary File Formats, Imports, Working with Imported Data.

FLUME – Apache Flume, Data Sources for FLUME, Components of FLUME Architecture.

UNIT –V:

Pig: Grunt, Comparison with Databases, Pig Latin, User Defined Functions, Data Processing Operators.

Spark: Installing steps, Distributed Datasets, Shared Variables, Anatomy of spark Job Run.

Scala: Environment Setup, Basic syntax, Data Types, Functions, Pattern Matching.

REFERENCE BOOKS:

1. Tom White, Hadoop: The Definitive Guide, 3rd Edition, O'Reilley, 2012.
2. V.K. Jain, Big Data & Hadoop, Khanna Publishing House, 2017. 2020-2021 118
3. P.J Sadalage and M. Fowler, NoSQL Distilled: A Brief Guide to the Emerging World ofPolyglot Persistence, Addison-Wesley Professional, 2012.
4. Eric Sammer, Hadoop Operations, O‟Reilley, 2012.
5. Lars George, HBase: The Definitive Guide, O'Reilley, 2011.

ONLINE RESOURCES:

1. https://www.tutorialspoint.com/hadoop/index.htm
2. https://www.tutorialspoint.com/hive/index.htm
3. https://www.tutorialspoint.com/hbase/index.htm
4. https://www.tutorialspoint.com/apache_pig/index.htm
5. https://www.tutorialspoint.com/scala/index.htm

COURSE OUTCOMES:

1. Understands the basics of big data and its real time examples.
2. Describe the design of HDFS and Hadoop I/O.
3. Demonstrate the Hadoop architecture and implementation of MapReduce Application.
4. Understand Hadoop related database tools such as NoSQL, HBase and Hive.
5. Design Pig Scripts for Big Data Applications.